

Arun Murugappan I

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Education

PES University, BTech in Electronics and Communications Aug 2022 – Jun 2026

- GPA: 7.93/10 upto sem 7
- **Coursework:** High Performance Computing, Computer Aided Digital Design, Computer Organization and Design, Digital VLSI Design, VLSI Circuit Testing and Testability, Cryptography,Wireless Mobile Systems,Quantum Computing.

Experience

NoC Verification Intern, InCore Semiconductors Jan 2026 – Jun 2026

- Enhanced an AMBA AHB/APB verification framework using Embedded UVM (D language) by rewriting golden reference model, scoreboard, sequences, drivers, and monitors
- Validated AMBA interconnects including AHB2AHB, AHB2APB, and AHB2AXI4 bridges, covering multiple data width configurations and protocol interactions.
- Worked on a scalable, design-independent verification framework enabling reuse across multiple DUTs

Teaching Assistant, PES University Aug 2024 – Dec 2024
Computer Aided Digital Design, under Prof. Vinay Reddy

- Assisted in conducting lab sessions and tutorials on RTL design, simulation, and FPGA implementation using Xilinx Vivado
- Assisted in lab sessions, clarified conceptual doubts, and provided hands-on support for assignments and projects

Teaching Assistant, CHIPS PES University June 2024 – July 2024

- Conducted hands-on lab sessions on PYNQ FPGA development boards and Zynq SoC architecture, covering hardware-software co-design concepts.

Student Researcher, Quanaad Lab PES University Jan 2023 – Dec 2024

- Fabrication of Quantum dots doped Poly-Aniline(PANI) and Poly-ethylenedioxythiophene(PEDOT) Nano-Films

Projects

Dynamic Partial Reconfiguration on FPGA Nov 2024 – Dec 2025
CHIPS, PES University

- Run-Time Adaptive Reconfiguration of RISC-V Processor's Co-Processor for resource optimization Using Xilinx Vivado DFX

RISCV SoC (RV32IM) June 2025 – Ongoing
Personal Project, PES University

- Designed a RISC-V (RV32IM) Core in Verilog with FPGA validation and SoC integration including UART, SPI peripherals.

FIFO (First-In-First-Out Memory) | RTL to GDSII Nov 2024 - Dec 2024
Personal Project, PES University

- Implemented a complete RTL-to-GDSII flow using Cadence toolchain with SCL180 PDK
- Performed logic synthesis, floorplanning, placement and routing, timing closure, and DRC/LVS checks to achieve a manufacturable layout.

PostGres SQL Acceleration using Alveo Cards June 2024 – March 2025

- Design of a database accelerator on the Alveo U50 card using Vitis Libraries.

AES Encryption/Decryption Co-Design on PYNQ FPGA Nov 2023 – Dec 2024

- Implemented AES encryption/decryption on PYNQ-Z2 with hardware-accelerated core and software based S-box/key generation.

Publication

Nanoclay-Based Conductive EMI Shielding Silver Decorated PANI: Developed silver-decorated polyaniline (Ag-PANI) nanocomposites enhanced with nanoclay via in-situ polymerization.

Technical Skills

Languages: SystemVerilog, Verilog, C, C++ , D, Python, Assembly

Tools: Xilinx (Vivado, Vitis), Cadence (Xcelium, Genus, Innovus, Virtuoso, Tempus),LTspice, Wireshark, RPiES

Scholarships

CNR Rao Scholarship *PES University, Bengaluru*
Merit Certificate + 40 Tuition Fee Refund for top 20 of the Students based on their SGPA.

Leadership & Extracurricular

Club Lead, Team Vegavath Racing Club

Sept 2022 – Dec 2024

PES University, Bengaluru

Bootstrap 2k24, Dept of Mechanical Engineering

Aug 2024

PES University, Bengaluru